# Patascore - NCBA Presentation




pezesha

www.pezesha.com
www.patascore.com

# Patascore, our proprietary credit scoring technology offers 85% balanced accuracy and less than 5% NPLs

**INPUTS**

DATA
- Transactional
- Bank statement
- Psychometric
- Geo location
- Credit Bureau
- Mobile money
- Demographic
- Business

Scoring

Extraction Module

Business Rules

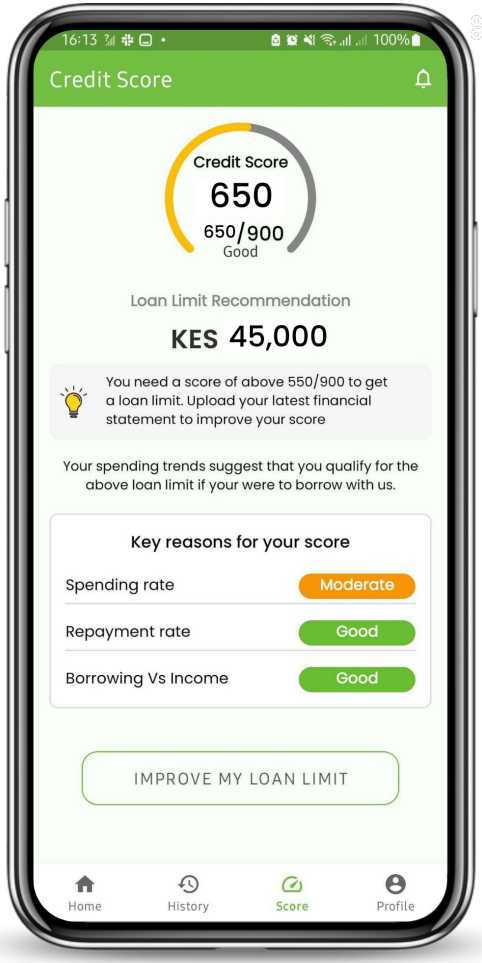
Fraud Detection

Patascore Machine Learning Modules

API

Real time

**SCORES, LIMITS, REASONS AND OFFERS**



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# #1

# Profile filling

## Personal details

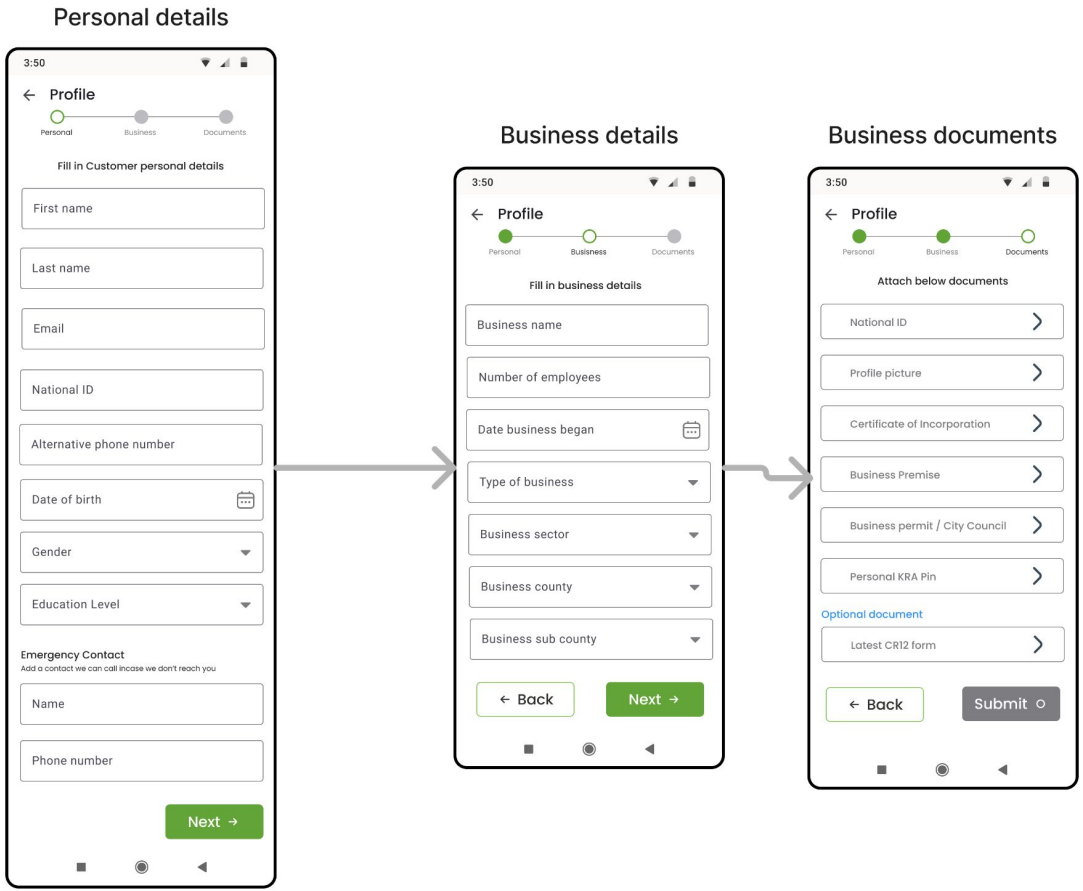
- Names
- email
- national ID
- date of birth
- Alternative phone number
- Emergency contact

## Business details

- Business name
- no. employees
- Date business began
- business sector
- Business county & sub-county

## Business documents

- National ID
- Profile picture
- ,certificate of incorporation
- Business premise
- Personal KRA pin
- CR12 document

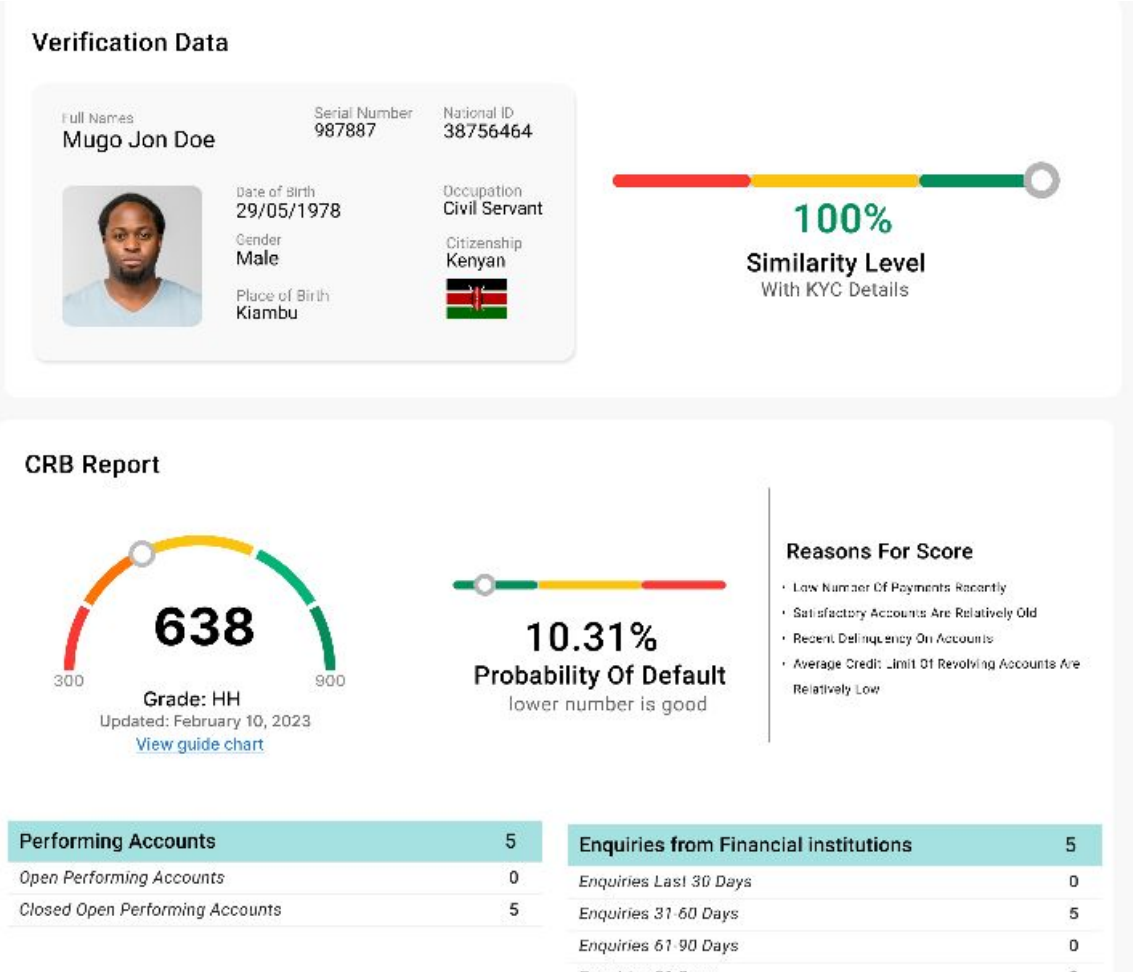


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#2

# CRB and IPRS Checks

With the combination of verification and credit check for customers, partners can give out loans with the level of credibility for an individual customer.



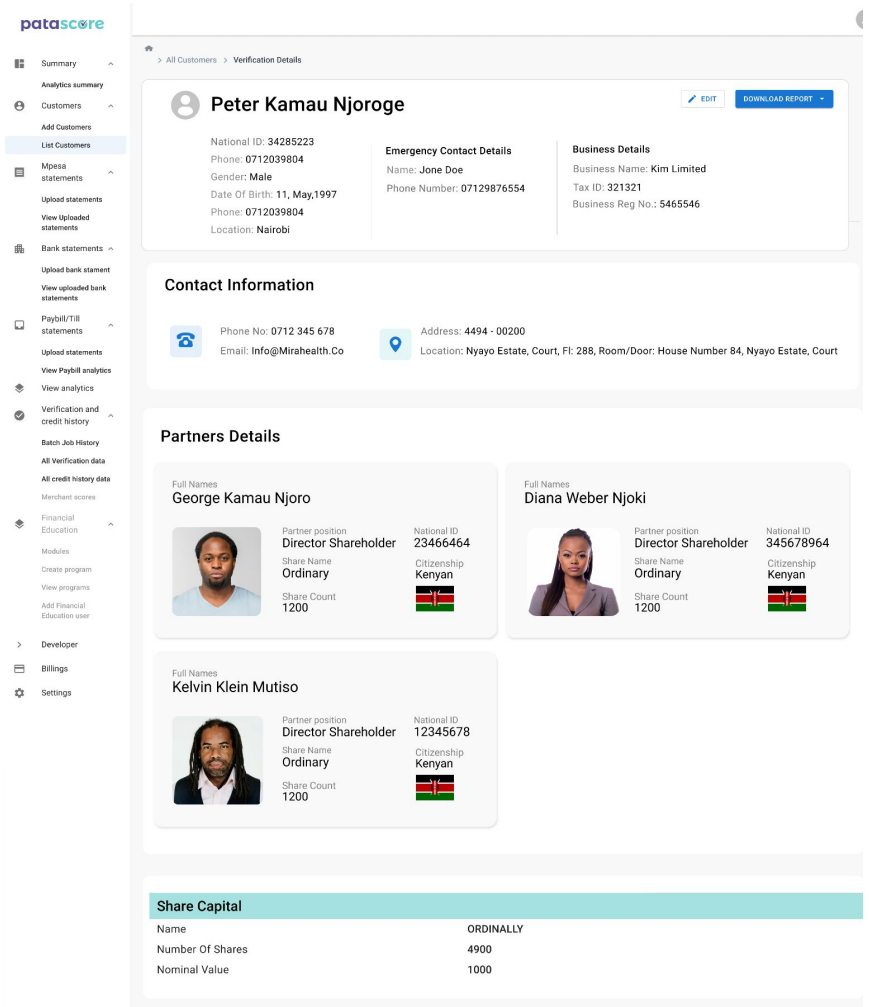
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# #3

## BRS checks

Full business registry checks on company profile, shareholders/partners, business contact information and registration history.

Patascore's BRS gives lenders real-time access to accurate and verified business information. This expedites the lending process and ensures lenders can make well-informed decisions based on a business's track record and credit history.



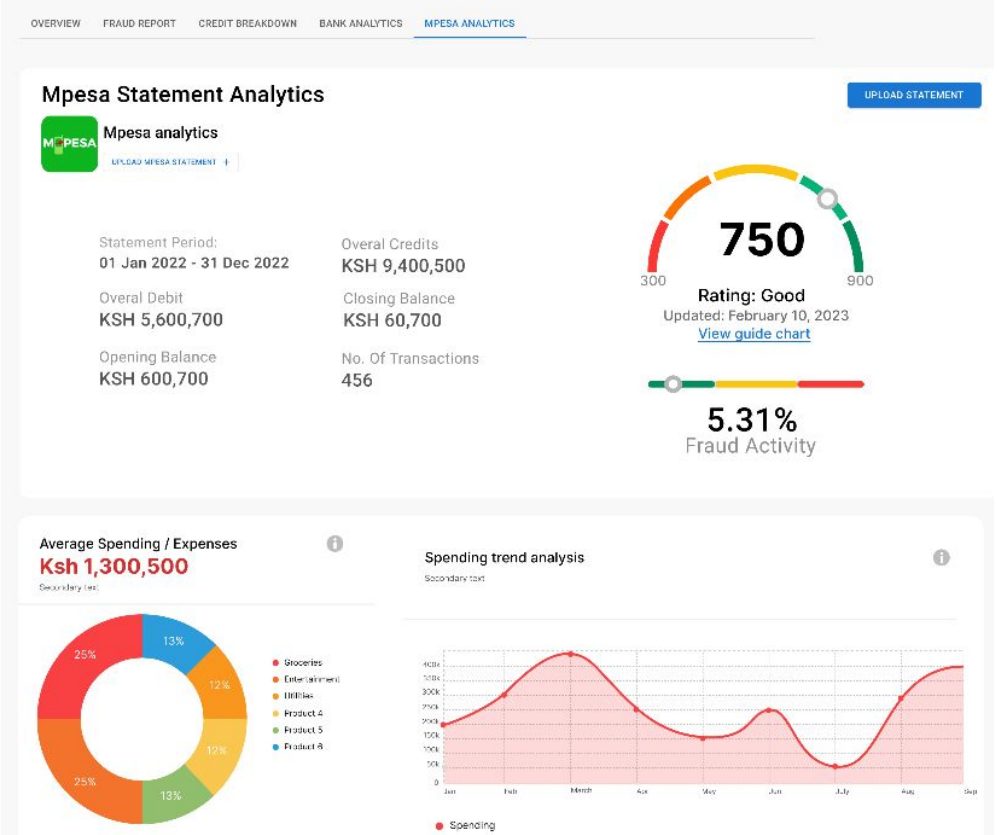
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# #4

# MPESA statement analytics

After Registering Customers, users can upload their Mpesa, Paybill , till for risk analytics credit scoring.

- *Easy upload process*
- *Conversion of statements to downloadable editable CSV documents*
- *Data converted to easy to read graphical visuals*



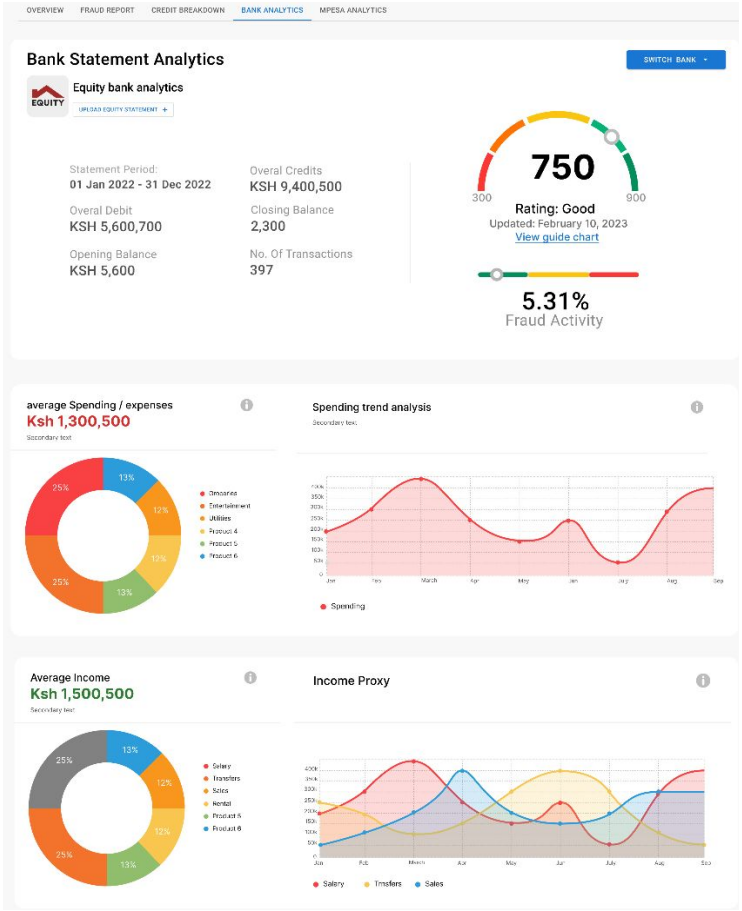
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# #5 (optional)

# Bank statement analytics

After Registering Customers, users can upload their bank statements for credit scoring.

- *Easy upload process*
- *Conversion of statements to downloadable editable CSV documents*
- *Data converted to easy to read graphical visuals*



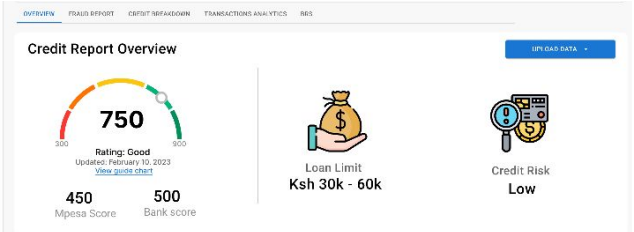
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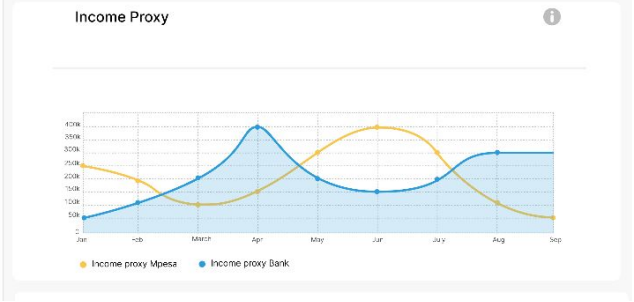
# #6

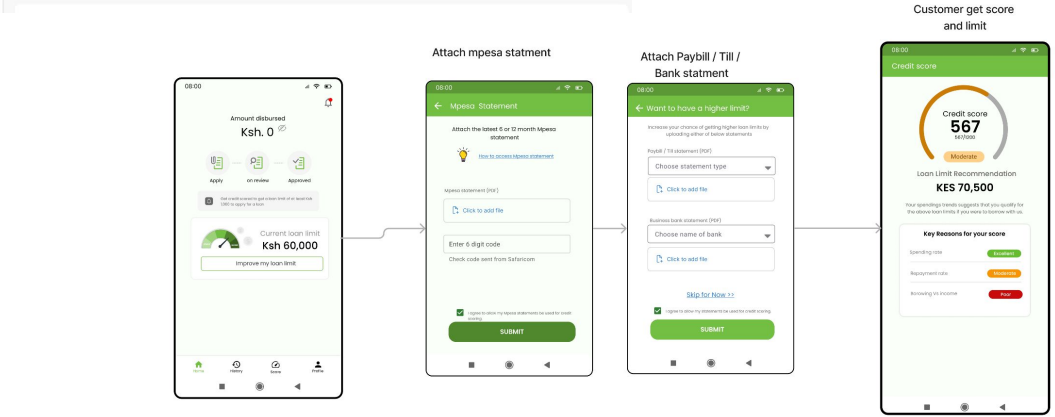
# Credit Scoring

CRB data, transactional information, customer credit history and financial statements are combined to give an overall credit score and a breakdown of a customer's income and risk profile.

- Get loan limit recommendations
- Overview analytics of the score
- Graphical presentations
- Mpesa , Till/paybill and bank analytics



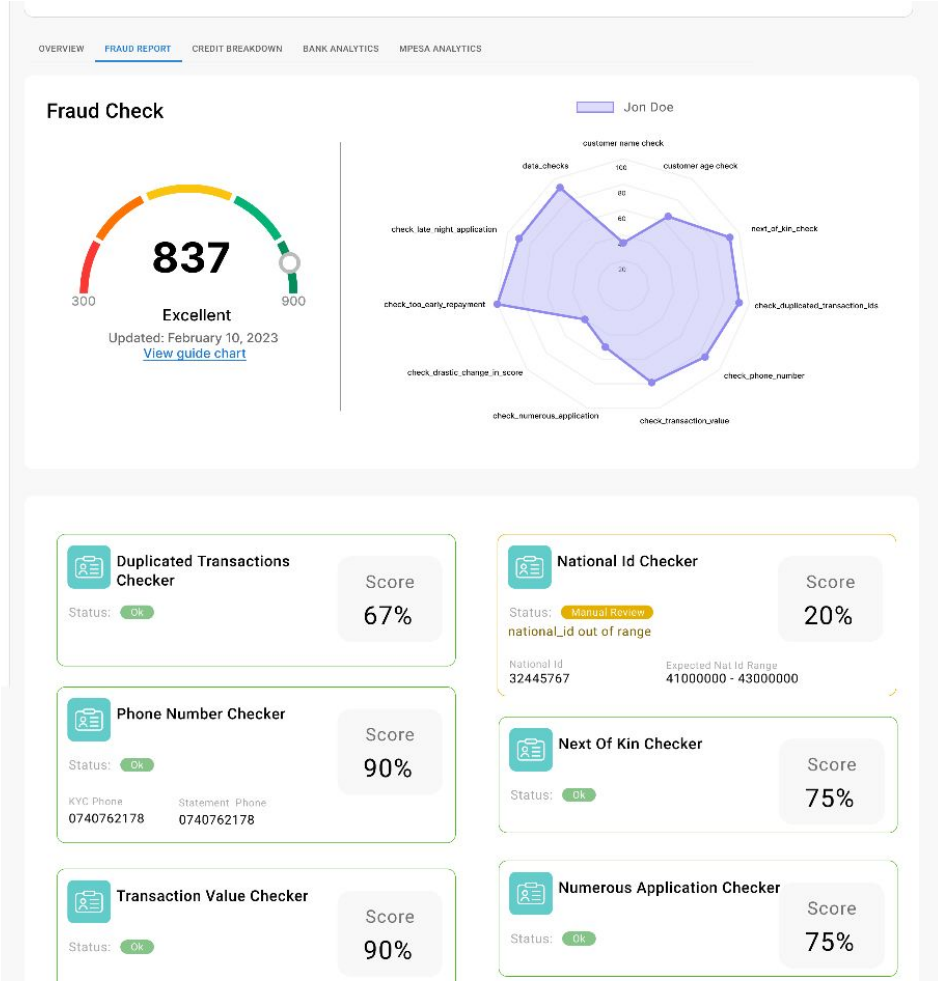
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# #7

# Fraud Detection

Patascore offers comprehensive fraud detection and risk management services. Our advanced algorithms are able to detect suspicious transactions, inconsistent relationships and similarity between customer datasets to ensure fraud risks are mitigated.

- Similarity levels
- Reason types
- Suspicious transactions



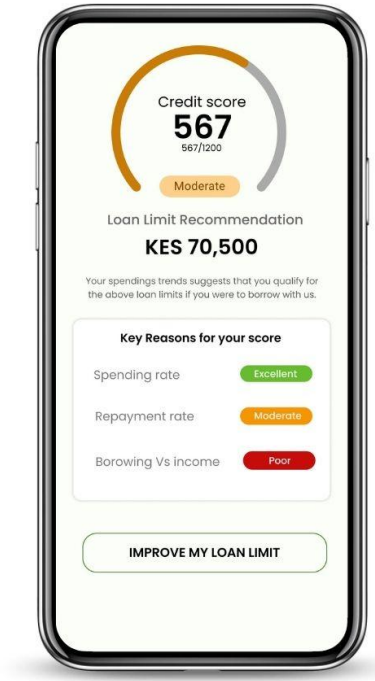
# Credit scoring approach and expected performance

**Data Binning and Weight of Evidence** → **Logistic Regression** → **Credit Scoring** → **Limit Assignment**

Pezesha uses a **combination of logistic regression and machine learning** models, combined with a weight of evidence modular approach to credit score MSME businesses.

Average model performance is currently **80% for new customers** and **90% for repeat customers***.

Pezesha's models have delivered a default rate of less than 5% and a PAR 30 less than 10% since inception.




*As measured by the F1 score, which captures the correct identification of bad customers, whilst minimizing false positives*

** Some anchor partner portfolios are more 'mature' and therefore offer better model performance than early stage partnerships*

***PAR-30 is defined by the value of loans that is outstanding by at least 30 days from the total outstanding loan value*

# Credit Scoring Modules

Credit scorecards are broken down into **'modules'** - each module reflects a segment of a scorecard. Before passing through the scoring modules, KYC checks, eligibility requirements, business rules and fraud checks are applied. **Modules can be modified, added and adjusted** depending on the product, segment or customer type.

Step 1: **Rules Engine** - this is where all requirements (data, KYC, fraud etc.) are checked

- KYC Module
- Fraud Module
- Gating Rules

Step 2: **Credit Scoring Engine -** modules applied to customer data

- Mobile Money Module
- \+ Transaction Data Module
- \+ CRB Module
- \+ Macro-economic Module
- \+ Credit history Module

Step 3: **Additional Modules (optional)**

- \+ Bank statement Module
- \+ Custom Made Module(s)

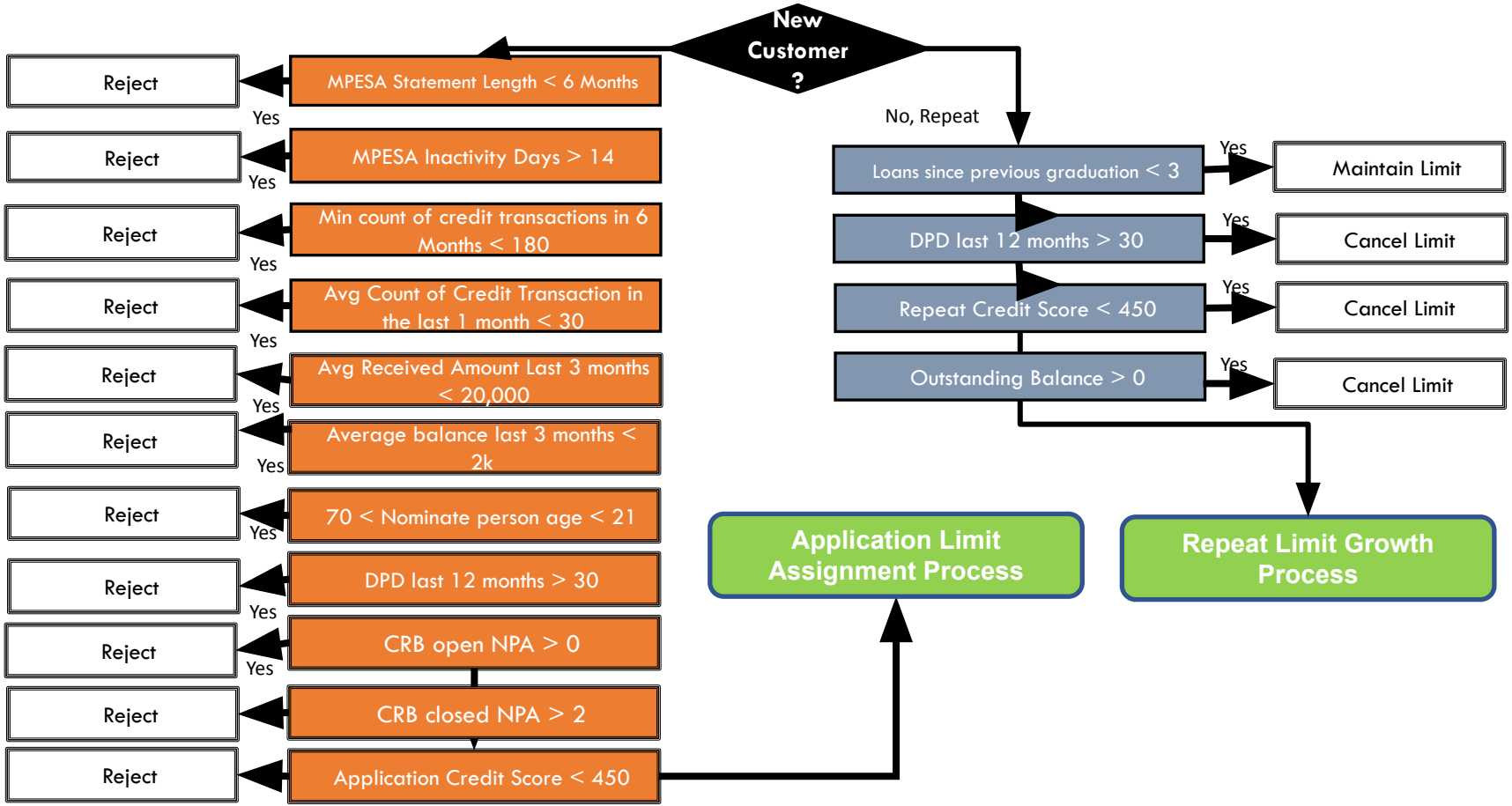
Apply weights and transformations




Step 4: **Loan Offer and Approval**

- Low Risk (score > 700): Limit Assignment and Offers
- Medium Risk (600 < score < 699): Limit Assignment and Offers
- High Risk (500 < score < 599): Limit Assignment and Offers

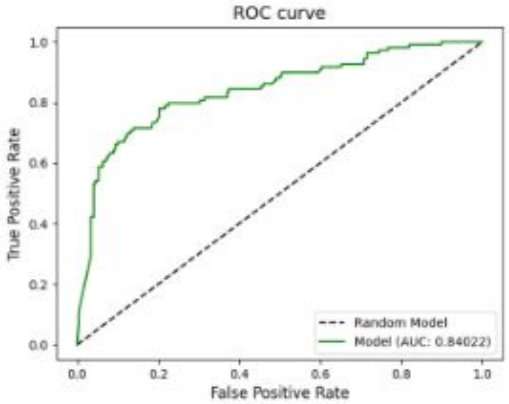
# Eligibility Criteria



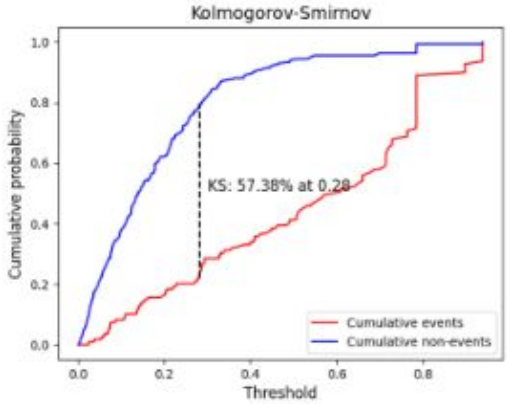
# Application Scorecard Performance

Scorecard artifacts for applicant (first time customer scorecard). The model(s) show high competency in separating good vs bad customers using primary transactional, CRB, KYC and behavioural data sources.

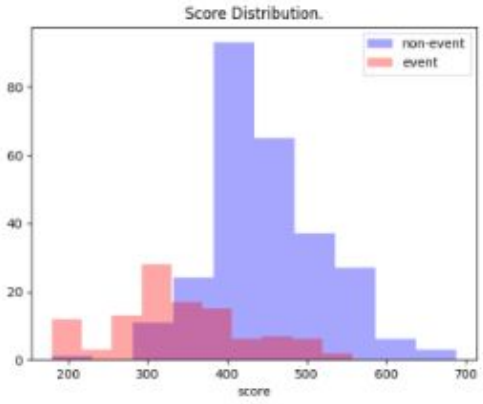
Average performance is 80% (as measured by the F1-score*).




(a) ROC Curve




(b) KS Plot



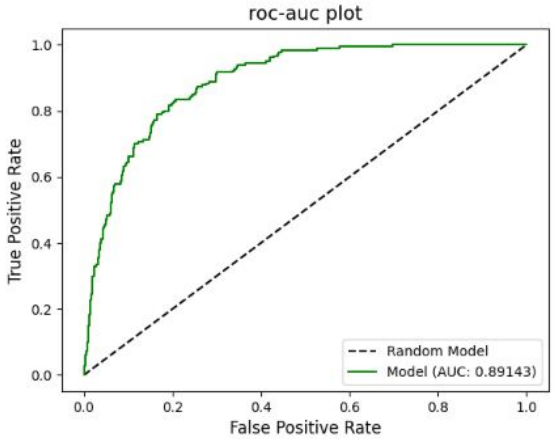

(c) Score Distribution

*The F1 score is a measure used to evaluate how well a classification model can correctly identify positive instances (i.e. defaults) while minimizing false positives.

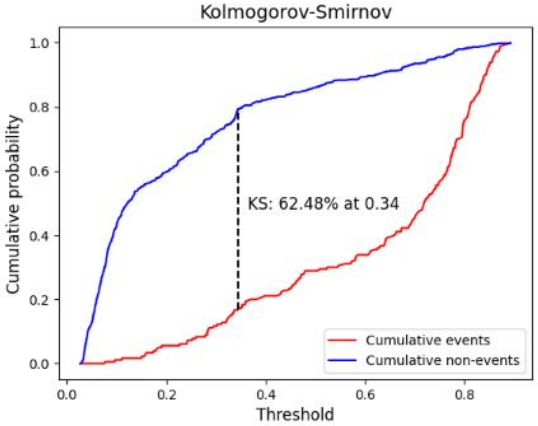
# Behavioural Scorecard Performance

Scorecard artifacts for behavioural (repeat customer scorecard). The model(s) show high competency in separating good vs bad customers using behavioral data sources.

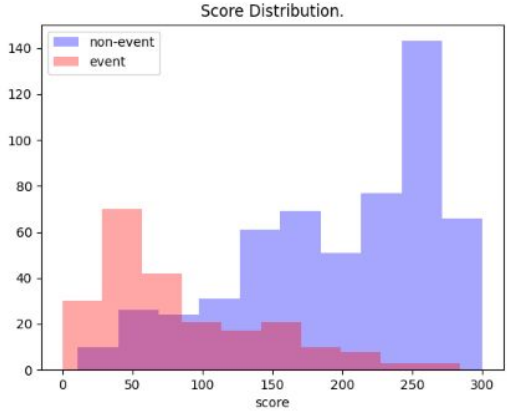
Average performance is 90% (as measured by the F1-score*).




(a) ROC Curve




(b) KS Plot



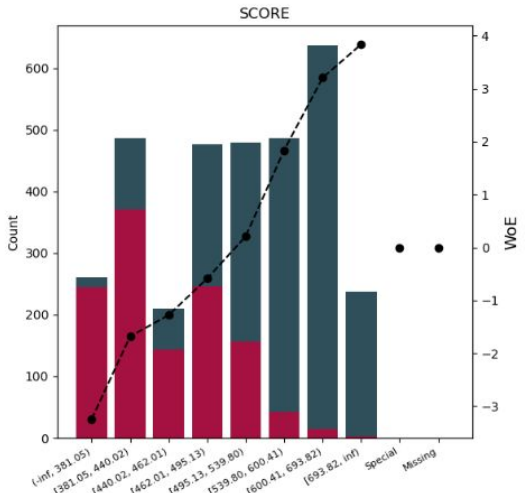

(c) Score Distribution

*The F1 score is a measure used to evaluate how well a classification model can correctly identify positive instances (i.e. defaults) while minimizing false positives.

# Gains Table

Customers to be **restricted to > 500 points to be fundable** during the pilot phase. This limits the number of high risk customers. From Pezesha's experience, customers in the medium - low risk buckets ensure a default rate of no more than 5%. Subject to portfolio performance score thresholds can be dialled up / down*.

| Bin | Count | Count (%) | Non-event | Event | Event rate | WoE | IV |
|---|---|---|---|---|---|---|---|
| (-inf, 381.05) | 260 | 0 | 16 | 244 | 0.9 | (3.2) | 0.6 |
| [381.05, 440.02) | 486 | 0 | 116 | 370 | 0.8 | (1.7) | 0.4 |
| [440.02, 462.01) | 210 | 0 | 67 | 143 | 0.7 | (1.3) | 0.1 |
| [462.01, 495.13) | 476 | 0 | 230 | 246 | 0.5 | (0.6) | 0.1 |
| [495.13, 539.80) | 479 | 0 | 323 | 156 | 0.3 | 0.2 | 0.0 |
| [539.80, 600.41) | 486 | 0 | 444 | 42 | 0.1 | 1.8 | 0.3 |
| [600.41, 693.82) | 637 | 0 | 622 | 15 | 0.0 | 3.2 | 0.9 |
| [693.82, inf) | 237 | 0 | 234 | 3 | 0.0 | 3.8 | 0.4 |



# Graduation Policy

Merchant to be subjected to repeat limit assessment after every 3 loans.
Subject to the repeat limit exclusion rules, the limits will be determined using the framework below;

| Graduation Matrix | current limit < 15k | 15k <= current limit < 25k | 25k <= current limit < 50k | 50k <= current limit |
|---|---|---|---|---|
| **Low Risk Credit Score >= 500** | Current Limit Growth Rate = 8% | Current Limit Growth Rate = 6% | Current Limit Growth Rate = 4% | Current Limit Growth Rate = 4% |
| **Medium Risk 390 <= Credit Score <= 499** | Current Limit Growth Rate = 5% | Current Limit Growth Rate = 3% | Current Limit Growth Rate = 2% | Current Limit Growth Rate = 2% |
| **Higher Risk 340 < Credit Score <= 389** | Current Limit Growth Rate = 0% | Current Limit Growth Rate = 0% | Current Limit Growth Rate = 0% | Current Limit Growth Rate = 0% |

= **(1+Growth Rate) * Current Limit**

- Initial loan offers to be capped at 50% average monthly credited values (Sum(3MCREDITAVG))
- Customers can only graduate every 3 loan cycles, if the loan cycle is not a factor of 3, i.e. 1, 2, 4, 5 then the limit should be maintain at the same level

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# Thank you

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